



NATIONAL SENIOR CERTIFICATE EXAMINATION
NOVEMBER 2009

INFORMATION TECHNOLOGY: PAPER I
MARKING GUIDELINES

Time: 3 hours

180 marks

These marking guidelines were used as the basis for the official IEB marking session. They were prepared for use by examiners and sub-examiners, all of whom were required to attend a rigorous standardisation meeting to ensure that the guidelines were consistently and fairly interpreted and applied in the marking of candidates' scripts.

At standardisation meetings, decisions are taken regarding the allocation of marks in the interests of fairness to all candidates in the context of an entirely summative assessment.

The IEB will not enter into any discussions or correspondence about any marking guidelines. It is acknowledged that there may be different views about some matters of emphasis or detail in the guidelines, and different interpretations of the application thereof. Hence, the specific mark allocations have been omitted.

Please note that learners who provided alternate correct responses to those given in the marking guidelines will have been given full credit.

QUESTION 1

- 1.1 Eavesdropping occurs when a person outside a network detects a signal on a cable without piercing the cable, thereby gaining access to the data.
 - 1.2 Multitasking is a processing technique whereby it appears as if more than one process/program is running at the same time.
 - 1.3 An algorithm based method used to check whether an error occurred during data transfer .
 - 1.4 A keylogger is a program or a hardware device which detects and records a user's keystrokes on a computer keyboard.
 - 1.5 WIMAX is a long-range wireless networking standard.
 - 1.6 Use of computer applications where the application(s) and data are hosted on servers on the Internet .
 - 1.7 An RSS feed is content which is automatically downloaded from the Internet by means of a feed reader program which checks constantly for new content.
 - 1.8 A podcast is an audio or video file broadcast (distributed) to a user via the Internet.
 - 1.9 A router is a device which connects networks that use different protocols.
 - 1.10 RAID is a technology whereby data is stored across two or more hard disks, to improve security of the data.
- In some of the above, candidates might provide more technical answers than these given. Some flexibility in the marking should be allowed.*

20 marks**QUESTION 2**

2.1

TABLE 1	Travelmate	eMachine
The amount of Level 2 cache	3 MB	1 MB
The speed of the system bus	1066 MHz	533 MHz
The amount of RAM	4096 MB RAM (or: 4 GB RAM)	512 MB RAM

(6)

- 2.2 This can be used to identify the user when logging on, eliminating the need for a password. (2)

(Must allow eliminates need for password as only answer).

- 2.3 In the Emachine, all instructions are processed by one processor; in the Travelmate, with the dual core CPU, two sets of instructions can be processed at the same time, as there are two CPUs. This increases the overall speed at which instructions are processed, thus increasing the overall work performance of the computer. (3)
- (Also allow 'works faster' in place of 'two instructions processed at same time')*

2.4 Any FOUR of:

- Pipelining
- Hyperthreading
- Cache memory
- Reduced size of transistors on CPU
- Clock multiplication
- Increased clock speed
- Increase in register size
- Reduce die size
- Improved cooling methods
- Improvements in instruction set

(4)

2.5 The Travelmate's graphics card has far more dedicated memory and shared memory than that of the eMachine, so it can store more information re data for graphics display. The FSB/system bus is much faster, so data can be transferred faster between the CPU, RAM and the graphics card. The CPU is far more powerful, so instructions which must be processed in order to send information to the graphics card will be processed faster.

(6)

(Mark allocation: 2 per category)

21 marks

QUESTION 3

3.1

3.1.1 More than one correct answer – must be substantiated clearly.
Examples:

Thomas should get a cheap version of Windows, as it easier to learn than a free OS such as Linux.

OR: Thomas can get a free version of Linux and get someone to help him learn how to work within this operating system.

(2)

(Note: No marks if the justification given is not valid.)

3.1.2 One of: Open Office, StarOffice, Siag Office, Neo Office, KOffice, Thinkfree Office

(1)

... add other valid answers (they must be **free office suites**) .

3.2

3.2.1 Any two of: Windows Mobile, Android, Symbian, OSX (Apple iPhone), Palm OS.

(2)

3.2.2 During the synchronisation process contacts are updated on both the cellphone and the computer so that both have identical sets of contacts. A similar process takes place for other components such as messages, tasks, etc.

(2)

- 3.3 The spam blocking program checks to see if the message is spam;
the anti-virus checks the message for viruses;
the application program opens and displays the message for Nadia to read;
the operating system opens and controls the application program;
the BIOS receives instructions from the operating system and communicates directly with the hardware, such as the printer. (5)
- 3.4
- 3.4.1 eMachine, Microsoft Works (2)
- 3.4.2 Any two of:
Word processor
Spreadsheet
Presentation program
Database
Games
... add other valid answers (2)
- 3.4.3 Any one of:
utility software
communication software
DBMS
security software
programming languages
... add other valid answers (1)
- 3.5 any two of:
File management
secondary storage management
I/O device management
Security management
Command interpreter (2)
- 3.6 Virtual memory is created by using an area on the hard disk to place some of the content of RAM. Memory pages are swapped between RAM and virtual memory as needed. (3)

22 marks

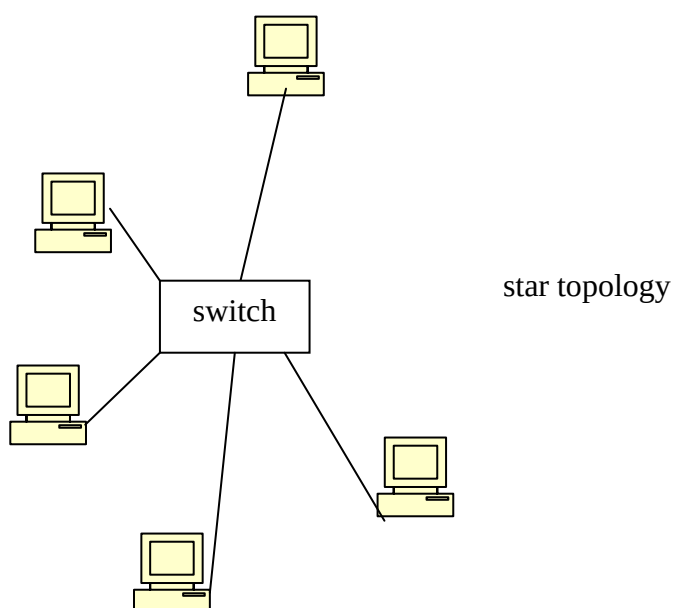
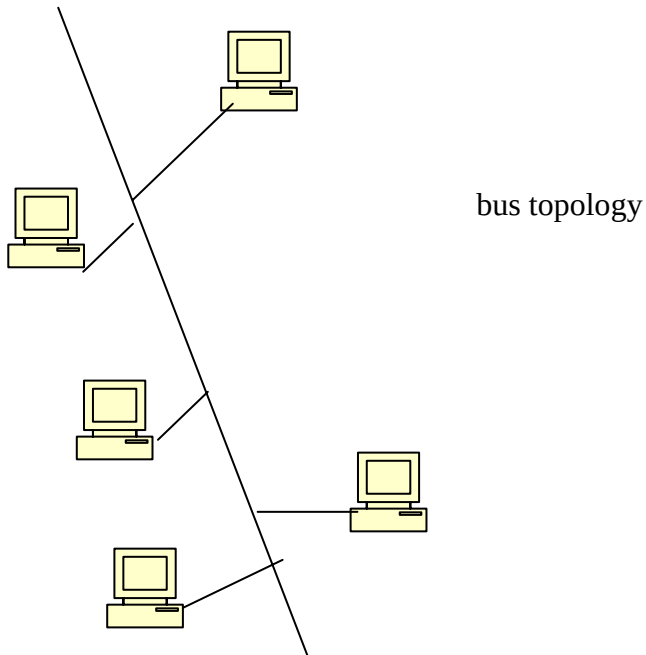
QUESTION 4

4.1

4.1.1 two of: UTP, STP, coaxial, fibre optic cables (2)

4.1.2 star, bus topologies (2)

4.1.3 two marks for sketch, none if it is not stated which topology this is (2)



- 4.2 All computers in the network are linked to the switch. The switch determines the destination of each network signal it receives, and forwards this signal to that destination.
OR
All computers in the network are linked to the switch. A switch makes sure that the communication goes directly to a specification destination in the network, without going to other computers. (3)
- 4.3
- 4.3.1 Ethernet LAN – she could plug her laptop LAN cable into an existing network.
Wireless LAN – she could connect to a WiFi hotspot such as at the airport or certain restaurants.
Bluetooth – she could send files between her laptop and her cellphone.
3G – she could use a 3G modem to access an ISP for Internet and e-mail activities when physically away from a wired network or WiFi hotspot. (4)
- 4.3.2 She could use Ethernet LAN or wireless LAN. (2)
- 4.3.3 Four marks for any combination of advantages/disadvantages
Disadvantages of peer-to-peer:
Less security than client-server
lack of central data storage
harder to control than client-server
Advantages of peer-to-peer:
Relatively cheap
does not need specialised knowledge for setting up (compared to a client-server network)
simpler to administer. (4)
- 4.4
- 4.4.1 Each workstation listens for traffic; if no traffic detected, the workstation sends a message. If the workstation detects a collision of this message with another message, it waits a random period of time and then tries to send the message again. (4)
- 4.4.2 Token ring. (1)

24 marks

QUESTION 5

- 5.1 When a person registers a web-based e-mail address, he/she receives an e-mail address from a web e-mail provider. Messages can be sent/received by logging onto the provider's e-mail site. Sent and received mail (usually) remains on the e-mail provider's server on the Internet. (3)
- 5.2 any two of:
gmail
webmail
hotmail
... add others (2)
- 5.3 any two of:
ensures that no data/communication enters through unauthorised ports
blocks communication from inside the network with unauthorised outside websites/
IP addresses
stops access to one's own computer/network by unauthorised users/software (2)
- 5.4
- 5.4.1 phishing (1)
- 5.4.2 She should not respond to this e-mail. One must never disclose any banking details/PIN numbers via e-mail. (2)
- 5.5
- 5.5.1 A document describing misconduct by a labourer contains sensitive information, and could be used incorrectly if the wrong person got hold of it. (1)
- 5.5.2 Encryption involves a key. The key is communicated between the sender and receiver of a document. The text of the document is scrambled using the key, and can only be decoded by using the key. (4)
- 5.6 Any two of: – search for web pages which:
contain specific phrases but not certain other phrases
have been on the web for a certain time period, (e.g. two weeks)
are of a certain file type, (e.g. PDF files only)
are written in a certain language only
have the keywords in the title of the page
appear in a specific domain only
... add valid answers (2)
- 5.7
- 5.7.1 A blog is a space on the Internet where an individual/company/ organisation can write about their experiences regarding a specific topic. (2)
- 5.7.2 Any one of:
she can receive feedback from others about what she is doing
the blog gives her an opportunity to make her work known to others
... add other valid answers (1)

5.8

5.8.1 Unfair use: examples of correct answers:

If Nadia published an article on casual labour on the Internet in which she uses text from another author, but does not acknowledge the author

OR

If Nadia publishes an article on casual labour for which she earns money and in which, she quotes another author at length, without the permission of that author

OR

Any example referring to use of material without acknowledgement or permission

(2)

5.8.2 Fair use: examples of correct answers:

If Nadia publishes an educational article about casual labour, in which she quotes and acknowledges another author

OR

If Nadia publishes an article showing what research has been done on casual labour, in which she quotes and acknowledges other authors

(2)

5.9

Any three of: – The owners of a specific website may state that:

they will not sell, rent or disclose private information to other parties

cookies are only used to improve the service offered by the website to the user, and not for illicit purposes

tracking of a visitor's movements will be used for the website company's own private purposes, and will not be disclosed to third parties

the company uses various security measures to protect users' information

changes to the privacy policy will be communicated to users, usually at a specific place on the website

(3)

5.10

The Great Health.com site is more reliable. Reasons: – any four of:

- it has links to reputable companies/sites
- it has recent information, such as the date 'October 2009' being mentioned
- it offers a privacy statement
- it offers a terms of use page
- it provides telephone numbers
- the Feel Great.com site appears to be 'pushy' about the products that it is selling
- the Feel Great site uses 'emotional language'

(5)

Note: Answers must refer to both sites.

5.11

The GoodHealth page is trying to avoid being inundated by spam.

(1)

33 marks

QUESTION 6

6.1 any two valid answers.

mark allocation: one mark per item, second mark if explained to some detail

Examples:

Efficient computerised systems at state hospitals would mean shorter queues when being admitted to hospital

Efficient computerised systems at Department of Labour means it is relatively quick to claim and receive UIF payments

Placing of ATMs in townships and rural communities allowing direct access to savings

Township and rural education through satellite computer education systems

Efficient computer systems used in issuing pensions – reduces long queues at payout offices

Expect **very**
varied answers

(4)

6.2

any three valid answers.

Examples – he can use a computer to:

create flyers to advertise his company

create a website to advertise his company

send emails to prospective customers

create a presentation to show to prospective customers

send podcasts

(3)

6.3

6.3.1

(6)

TABLE 2	Hardware knowledge and skills	Software knowledge and skills
Joey	Joey needs to know some basics, so that if a phone-in client's problem is hardware related (such as printing problems) she could advise the client to seek help regarding hardware.	Joey needs extensive knowledge of all Microsoft Office applications, and in addition to that, other software as well. Clients could phone in on any software created by Microsoft.
Michael	Michael needs to know hardware very well, as he is fixing computers on site.	Michael needs extensive software knowledge regarding operating systems and networking software, as many of the problems he encounters on site could be caused by software problems.
Thandeka	Thandeka needs hardly any hardware knowledge, as her daily teaching follows a set pattern regarding a software syllabus.	Thandeka needs to know Microsoft Word, Excel and PowerPoint very well.

6.3.2 (a) any two valid answers:

Examples:

Joey needs to learn Microsoft Word, Excel and PowerPoint in greater depth than she did before, as in her previous job she would not have to deal with every aspect of each program.

She may find it difficult to adjust to working with many people who have very little knowledge of working with a computer

(2)

(b) any two valid answers

Examples:

Thandeka needs to learn details of other Microsoft software besides Word, Excel and PowerPoint.

She may have a problem learning the telephone skills required in a call support centre.

People might phone in regarding tricky problems which Thandeka would not have encountered in her computer literacy training.

(2)

6.4 any two valid answers: the password should:

be 8 characters or longer

contain a mixture of digits and non-alphanumeric characters

should not consist of names of relatives, pets, etc.

should not be based on one's birthday, or any other personal details

add other valid answers...

(2)

6.5 any well substantiated answer:

examples:

Yes, they should block access, as employees will spend too much work time on social communication

Well-substantiated 'no' answers will also be accepted.

(2)

6.6 any two valid answers:

examples:

information about suspected criminals can be communicated swiftly across the network

a central database regarding criminals can be accessed instantly

general administration of police stations will improve as persons in managerial positions can communicate instructions swiftly

(4)

6.7 four marks allocated as follows:

if single argument (bank or John) with single justification: two marks

if arguments both ways with justification: full marks

Examples:

The bank could be to blame if:

their firewall protection is not adequate

their procedures are not secure enough (for example, accepting telephonic instructions to change something regarding a person's bank account)

John could be to blame if:

He does not update his anti-virus program regularly (required by the banks)

His computer's firewall protection is not adequate

He banked at an Internet cafe

He does not change his logon codes on a regular basis

He did not keep his PIN number secret

(4)

6.8 Any two valid answers:

Examples:

Computer systems help us analyse statistics about global warming

Computer systems help us predict tornadoes/hurricanes/tidal waves

Websites that help highlight issues, raise awareness, etc.

... add other answers

(2)

31 marks

QUESTION 7

7.1 Create separate fields for the street name and number, suburb, city, etc. (2)

7.2 To identify a record uniquely. (1)

7.3 No, this will not identify a record uniquely, as a particular combination of skillID and empID can appear more than once. (2)

7.4 two possible ways of giving the answer:
the table must meet the requirements of 1NF
there must be no partial dependencies
OR
There must be no partial dependencies
Plus: any one of the requirements of 1 NF:
No repeating groups
No data with multiple values
Must have a primary key (2)

7.5
7.5.1 update anomaly (1)
7.5.2 Mbemba's address will have to be changed in more than one record (1)

7.6 Purpose of normalisation:
To minimise redundancy and the potential for errors. (2)

7.7 macID (1)

7.8 SELECT machine, count(*) AS Totals
FROM tblMachinery
GROUP BY machine; (2)

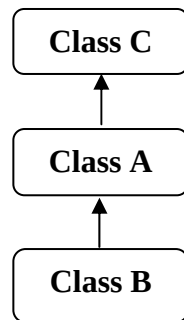
7.9
SELECT tblMachinery.machine, tblMachinery.operator, tblRentals.client,
tblRentals.date
FROM tblMachinery, tblRentals
WHERE tblMachinery.macID = tblRentals.macID;
(mark allocation: 1 mark for SELECT ... FROM ... WHERE structure
1 mark for correct selection of fields/tables following each of SELECT,
FROM, WHERE (4)

7.10

7.10.1 Class C.

(1)

7.10.2 – one mark for up arrow, two for correct placement of classes



(3)

7.11

7.11.1 space is set aside in RAM for the fields and methods of the class.
fields of the class are initialised to specific values.

(2)

7.11.2 a parameterised constructor receives value(s) from the statement which calls it.

(1)

7.11.3 any valid answers:

example:

an object such as Tutor is instantiated; the field 'ratePerHour' could be initialised to the value specified in the constructor's parameter.

(2)

7.12

Accessor methods receive requests to return the value stored in a field.

Mutator methods receive requests to change the value of a field.

(2)

29 marks**Total: 180 marks**